ESpeech: A Large-Scale, High-Quality Russian Speech Corpus for Text-to-Speech

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Abstract

High-quality, large-scale, and publicly available speech corpora are critical for advancing research in Text-to-Speech (TTS) synthesis. While numerous such datasets exist for English, the resources for the Russian language remain notably scarce and often suffer from limitations such as small size, limited domain variety, or insufficient quality control. To address this gap, we introduce ESpeech, a new large-scale Russian speech corpus comprising over 4700 hours of audio. The corpus is curated from distinct domains, including podcasts and educational webinars, providing a rich diversity of speaking styles. In addition to our large multi-speaker corpora, we also release four high-quality, single-speaker datasets. We present a sophisticated, multi-stage data processing pipeline designed to automatically extract high-quality speech segments from raw, "in-the-wild" audio. Our pipeline includes modules for source separation, speaker diarization, advanced VAD segmentation, and rigorous quality filtering based on a composite Mean Opinion Score (MOS).

1 Introduction

The performance of modern deep learning models for speech synthesis is heavily dependent on the volume and quality of the training data. For many languages, particularly English, the research community benefits from a wealth of large-scale corpora like LibriTTS [6]. These datasets have been instrumental in pushing the boundaries of what is possible in TTS, enabling the creation of natural and expressive synthetic voices.

However, the landscape for the Russian language is markedly different. Existing public corpora are often limited in several key aspects. Many are single-speaker datasets recorded in sterile, anechoic chambers, which, while clean, lack the prosodic and emotional diversity found in natural human speech. Other multi-speaker datasets compiled from public sources frequently suffer from inconsistent audio quality, poor transcription accuracy, and inadequate filtering, making them unsuitable for training state-of-the-art TTS models without significant manual effort. This scarcity of high-quality data has become a major bottleneck for Russian speech technology research.

To bridge this crucial gap, we present the ESpeech corpus, a new, extensive collection of Russian speech data. Our contribution is manifold: first, the corpus itself, which contains over 4700 hours of diverse audio, including large multi-speaker collections and four curated single-speaker datasets; and second, the fully automated pipeline we developed to create it. This pipeline is specifically designed to process challenging, real-world audio, transforming it into clean, well-segmented, and accurately transcribed samples ready for model training.

This paper details the construction of the ESpeech corpus. We first describe the data sources in Section 2. We then provide a comprehensive overview of our multi-stage processing pipeline in Section 3. Finally, we present the statistics of the final corpus in Section 4 and conclude with our findings and directions for future work.

2 Data Sources

The ESpeech corpus is built from several distinct and complementary sources to ensure a wide range of acoustic and stylistic diversity.

2.1 Espeech-podcasts

This sub-corpus is sourced from a wide variety of Russian-language podcasts publicly available on YouTube. The topics range from technology and science to entertainment and interviews. This source provides a rich collection of spontaneous, conversational speech with multiple speakers, diverse accents, and varying background conditions. The primary challenge with this data is its "in-the-wild" nature, often containing music, sound effects, and significant acoustic noise.

2.2 Espeech-webinars

This sub-corpus consists of recordings from educational webinars hosted by the Skolkovo online school. This data typically features a primary speaker (the lecturer) delivering a structured monologue, which is often clearer and more formal than podcast speech. However, it can also include interactions with students, audio from presentations, and other non-speech sounds. This source provides long-form, high-quality recordings of single speakers, which is highly valuable for TTS.

2.3 Single-Speaker Corpora

In addition to the large multi-speaker collections, we release four high-quality, single-speaker datasets. These were curated using the same automated pipeline from public sources, focusing on speakers with substantial amounts of publicly available audio content. These datasets are ideal for training single-speaker TTS models with distinct vocal characteristics. The released datasets are: **ESpeech-igm**, **ESpeech-buldjat**, **ESpeech-upvote**, and **ESpeech-tuchniyzhab**.

3 Data Processing Pipeline

To convert the raw audio files into a high-quality dataset, we developed a modular, automated pipeline. Each module performs a specific task, progressively refining the data. A diagram of the pipeline is shown in Figure 1.

3.1 Standardization

The raw speech data in the wild vary in encoding formats, sampling rates, etc. Therefore we convert all samples to a mono channel, and resample to 44.1 kHz. We set the sample width to 16-bit and adjust the target decibel level to -20 dBFS. The volume adjustment is constrained between -3 and 3 dB to avoid distortion. Finally, we normalize the waveform by dividing each sample by the maximum amplitude, ensuring values range between -1 and 1. These steps ensure a consistent data format for further processing.

3.2 Audio Quality Checker

The second stage after audio standardization is preliminary quality checking. For this, a mini-model is used that classifies audio into two classes: 'High' and 'Low'.

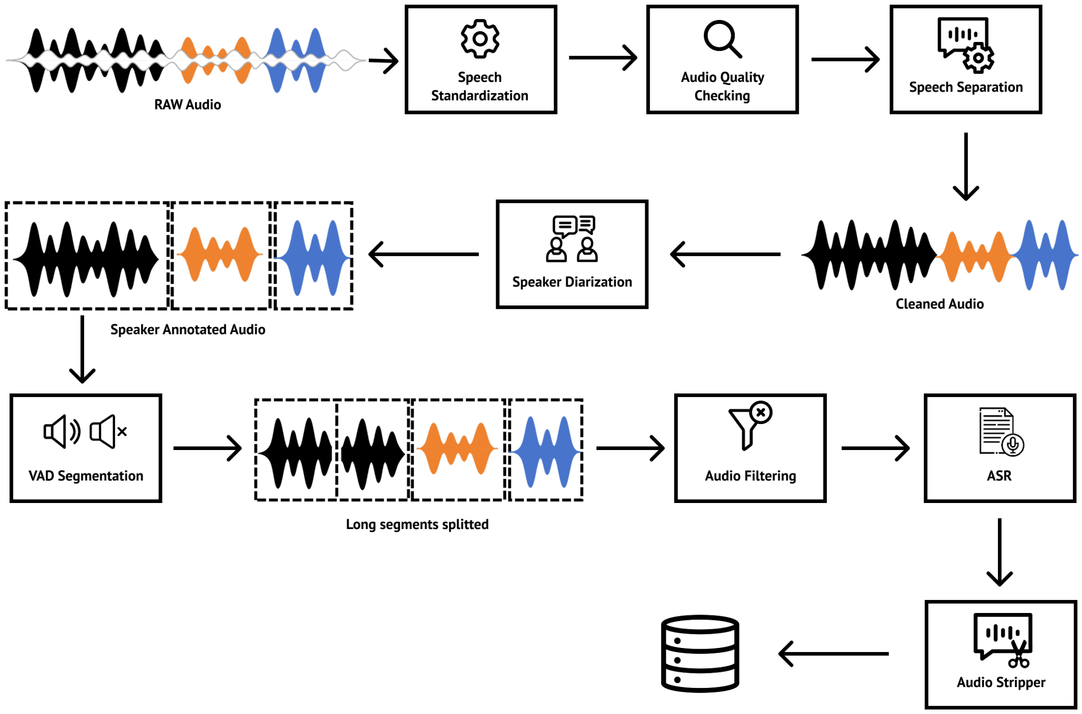

Figure 1: The ESpeech data processing pipeline.

3.3 Source Separation

To handle the "in-the-wild" nature of our sources, especially the podcasts, we employ a ClearerVoice [7] source separation model. This module is tasked with isolating the speech signal by removing or significantly attenuating background music, ambient noise, and other non-speech sounds. To optimize computational efficiency while maintaining quality, we distilled the original ClearerVoice model, reducing its size by half while preserving most of its performance capabilities.

Model	MOS Score
Raw audio	1.73
UVR (baseline)	2.42
ClearerVoice-original	2.97
ClearerVoice-distilled	2.88

Table 1: Mean Opinion Score (MOS) comparison of different audio processing approaches.

As shown in Table 1, our distilled model achieves a MOS score of 2.88, only 0.09 points lower than the original model (2.97), while providing significant computational savings through the 50% size reduction and speeding it up to 80 xRT.

3.4 Speaker Diarization

Once the speech signal is isolated, we use the PyAnnote toolkit [2] for speaker diarization. This process segments the audio stream and assigns a unique identity to each speaker. This is crucial for separating speech from different individuals in multi-speaker recordings, ensuring that each final audio sample belongs to a single speaker.

3.5 VAD Segmentation

The segments produced by diarization can still be very long, sometimes spanning several minutes. Such long segments are unsuitable for transcribing using Whisper [4]. We use SileroVAD [5] to further break down long segments (typically those exceeding 30 seconds) into smaller, more manageable chunks based on voice activity.

3.6 Overlap Filtering

Speaker diarization is not perfect and may occasionally produce segments where the speech of two speakers overlaps. To ensure each sample contains only one voice, we use a custom-trained audio classification model. This model analyzes each segment and discards those identified as containing overlapping speech.

3.7 MOS Filtering

A key innovation of our pipeline is a multi-faceted quality filtering step based on Mean Opinion Score (MOS) estimation. Using a model based on GigaAM audio encoder [3], we compute three distinct quality metrics for each segment:

- **NoiseMOS:** Measures the level of background noise.
- **ExpressiveMOS:** Estimates the prosodic and emotional richness of the speech.
- **ClarityMOS:** Assesses the intelligibility and clarity of the speech.

These three scores are then combined into a single, unified MOS score. Samples falling below a predefined threshold for this composite score are discarded, ensuring a high-quality final dataset.

3.8 ASR with WhisperX

For transcription and alignment, we use WhisperX [1], a framework for OpenAI's Whisper model that provides accurate word-level timestamps. Critically, we feed our pre-segmented audio chunks directly to WhisperX, bypassing its internal VAD.

3.9 AudioStripper

As a final cleaning step, we apply a model we call AudioStripper. VAD can sometimes make small errors, leaving short periods of silence, breath sounds, or residual noise at the beginning or end of a segment. This module precisely trims these artifacts, resulting in clean, tightly-cropped audio samples perfect for TTS training.

4 Dataset Statistics

The final ESpeech corpus is divided into the sub-corpora described in Section 2. The key statistics for the large multi-speaker corpora are presented in Table 2. The podcast data, being more diverse and conversational, achieves a slightly higher ExpressiveMOS, while the total volume is significantly larger. The webinar data, while smaller, exhibits high clarity.

Table 3 details the statistics for the four released single-speaker datasets. These corpora demonstrate high overall quality, with particularly strong scores in noise reduction (NoiseMOS).

Table 2: Key statistics for the multi-speaker ESpeech corpora.

Metric	Espeech-webinars	Espeech-podcasts
Total Hours	850	3340
Average MOS	3.00	3.10
NoiseMOS	4.10	4.30
ExpressiveMOS	2.50	2.80
ClarityMOS	3.50	3.80

Table 3: Statistics for the single-speaker ESpeech sub-corpora.

Metric	ESpeech-igm	ESpeech-buldjat	ESpeech-upvote	ESpeech-tuchniyzhab
Total Hours	220	54	296	306
Average MOS	3.30	3.14	3.10	3.00
NoiseMOS	4.36	4.36	4.50	4.185
ExpressiveMOS	2.80	2.80	2.60	2.60
ClarityMOS	3.40	3.40	3.30	3.30

5 Conclusion

We have presented ESpeech, a new large-scale Russian speech corpus containing over 4700 hours of high-quality, transcribed audio. We have also detailed the automated pipeline developed for its creation, which effectively processes "in-the-wild" audio through a series of cleaning, segmentation, and filtering stages. By leveraging diverse domains and employing a rigorous MOS-based quality control system, the ESpeech corpus provides a rich resource for the research community. The release includes both large-scale multi-speaker corpora and four distinct, ready-to-use single-speaker datasets. We believe this comprehensive dataset will significantly lower the barrier to entry for developing high-quality Russian TTS systems and will serve as a valuable benchmark for future research in Russian speech processing.

All components of the ESpeech corpus are publicly available on Hugging Face: https://huggingface.co/datase

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